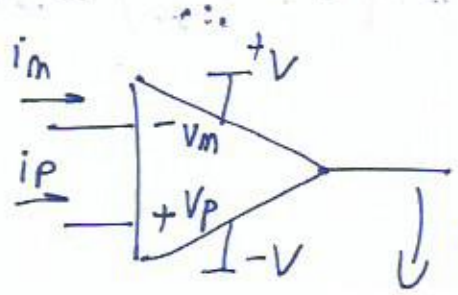
FICHA nº 7

22/ AMP-OP ⇒ REALIMENTAÇÃO NEGATIVA

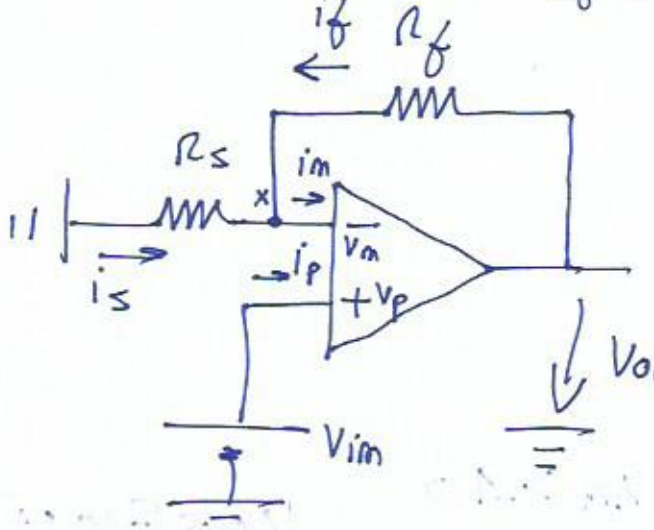
CONDIÇÕES:
- $i_m \cong i_p \cong 0$
- $V_m \cong V_p$



$R_s = 10k$ $\quad V^+ = 15V$

$R_f = 5k$ $\quad V^- = -15V$

a)



· LEI DOS NÓS:

no (x): $i_s + i_f = i_m = 0$

· LEI DE OHM:

$$i_s = \frac{0 - V_x}{R_s}$$

$$i_f = \frac{V_{out} - V_x}{R_f}$$

MAS $\Rightarrow \begin{cases} V_p \cong V_m \\ V_p = V_{in} \end{cases} \Rightarrow V_x = V_{in}$

$V_x = V_{in}$

$$\begin{cases} i_s = \dfrac{0 - V_{in}}{R_s} \\ i_f = \dfrac{V_{out} - V_{in}}{R_f} \end{cases}$$

(no x) $i_s + i_f = 0 \Rightarrow \dfrac{0 - V_{in}}{R_s} + \dfrac{V_{out} - V_{in}}{R_f} = 0$

$$\frac{V_{out}}{R_f} = \frac{V_{in}}{R_s} + \frac{V_{in}}{R_f} \Rightarrow \boxed{V_{out} = V_{in} \times \left(1 + \frac{R_f}{R_s}\right)}$$

b)

$V_{im} = 5^-$

$V_{our} = V_{im} \times \left(1 + \frac{R_f}{R_5}\right) = 5^- \times \left(1 + \frac{5k}{10k}\right) = 5^- \times (1+0{,}5) = 7{,}5V$

c) SIMULAR PSPICE

d) GAMA VALORES V_{im}:

$V_{our} \in [-15, 15]$

$|V_{im} \times (2+0.5)| \leq 15$

$V_{im} \times 1.5 \in [-15, 15]$

$$\begin{cases} V_{im} \times 1.5 > -15 \\ V_{im} \times 1.5 < +15 \end{cases} \Rightarrow \begin{cases} V_{im} > -10 \\ V_{im} < 10 \end{cases}$$

$V_{im} \in [-10, 10]$

23)

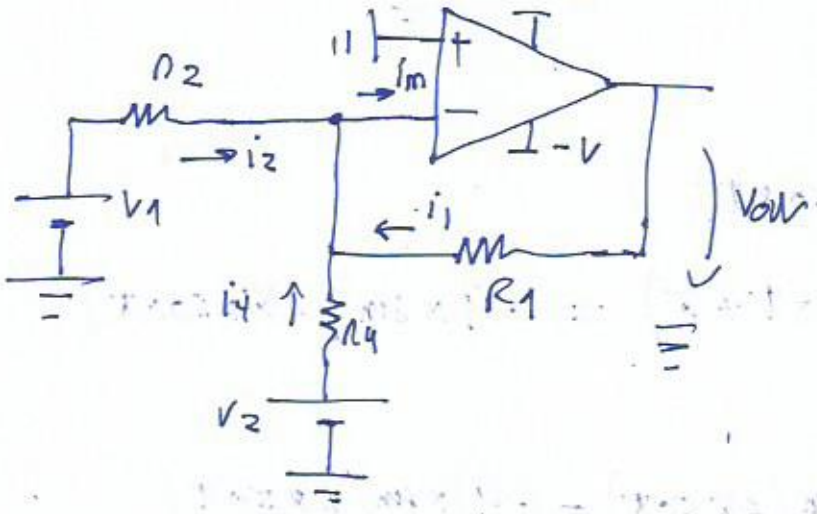
$V^+ = 15$

$V^- = -15$

$R_1 = 2k$

$R_2 = 2k$

$R_4 = 2k$



c) $V_{out} = ?$ $\left\{ \begin{array}{l} V_1 = 1V \\ V_2 = 2V \end{array} \right.$

$V_p = 0 \Rightarrow V_m = V_p = 0$

$i_2 + i_1 + i_4 + i_m = 0$

$$i_2 = \frac{V_1 - V_m}{R_2}$$

$$i_4 = \frac{V_2 - V_m}{R_4}$$

$$i_1 = \frac{V_{out} - V_m}{R_1}$$

$(V_m = 0)$

$$\frac{V_1 - 0}{R_2} + \frac{V_{out} - 0}{R_1} + \frac{V_2 - 0}{R_4} = 0$$

$$\frac{V_1}{R_2} + \frac{V_{out}}{R_1} + \frac{V_2}{R_4} = 0$$

$$\frac{V_1}{2k} + \frac{V_{out}}{2k} + \frac{V_2}{2k} = 0$$

$$V_{out} = -V_1 - V_2$$

$V_1 = 1, V_2 = 2 \Rightarrow V_{out} = -3V$

b/ SIMULAR PSPICE

c) $V_2 = -2$

$V_{OUT} = ?$

$V_{OUT} = -V_1 - V_2 = -1 - (-2) = +1\ V$

d)

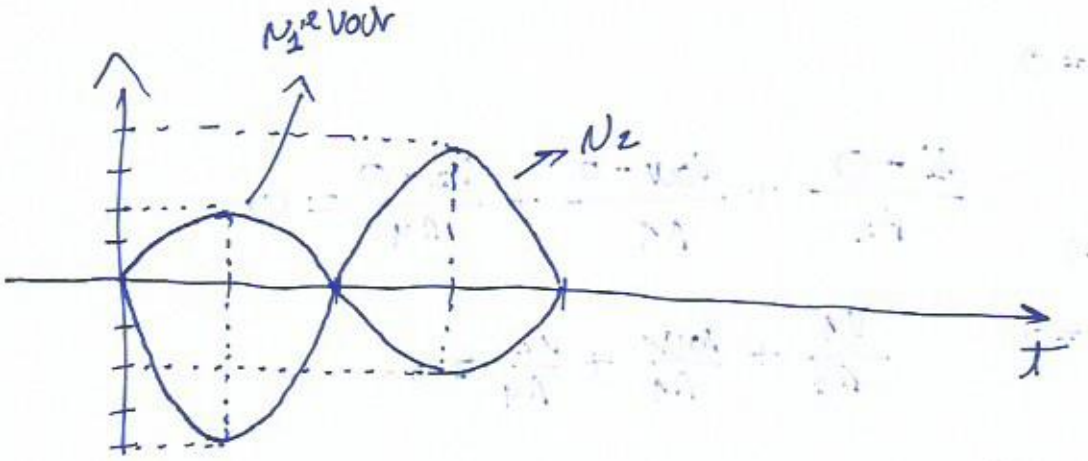
$V_1 = 2 \times \text{sen}(2 \times \pi \times 50 \times t)$

$V_2 = 4 \times \text{sen}(2 \times \pi \times 50 \times t + \pi) = -4 \times \text{sen}(2\pi \times 50 \times t)$

$V_{OUT} = -V_1 - V_2 = -2 \times \text{sen}(2\pi 50t) - (-4) \times \text{sen}(2\pi 50t)$

$= -2 \times \text{sen}(2\pi 50t) + 4 \times \text{sen}(2\pi 50t)$

$= 2 \times \text{sen}(2\pi 50t)$



e)

f) SIMULACIÓN PSPICE

g) $V_1 = 5$

$V_2 = ?$

$$V_{OUT} = -V_1 - V_2 = -5 - V_2$$

$$-15 < V_{OUT} < 15 \Rightarrow \begin{cases} \cdot\ V_{OUT} < 15 \Rightarrow -5 - V_2 < 15 \\ \cdot\ V_{OUT} > -15 \Rightarrow -5 - V_2 > -15 \end{cases}$$

$$-5 - V_2 < 15 \Rightarrow 5 + V_2 > -15 \Rightarrow V_2 > -20$$

$$-5 - V_2 > -15 \Rightarrow 5 + V_2 < 15 \Rightarrow V_2 < 10$$

$$V_2 \in [-20, 10]$$

h) $V_1 = ?$

$V_2 = ?$

ENTRADA DIGITAL $V_1\ V_2$ $\Rightarrow$ SAÍDA ANALÓGICA

D_1	D_2	V_{OUT}
0	0	0
0	1	-1
1	0	-2
1	1	-3

ANÁLISE DO CIRCUITO:

$$\frac{V_1 - 0}{R_2} + \frac{V_2 - 0}{R_4} + \frac{V_{OUT} - 0}{R_1} = 0$$

$$\frac{V_{OUT}}{R_1} = \frac{-V_1}{R_2} - \frac{V_2}{R_4}$$

ESPECIFICAÇÕES:

$\Downarrow$

CÁLCULO DE

$R_1, R_2, R_4 = ?$

$$\frac{V_{OUT}}{R1} = \frac{-V_1}{R_2} - \frac{V_2}{R4}$$

1. $V_1 = 0$, $V_2 = 0$ $\Rightarrow V_{OUT} = 0 \Rightarrow \frac{V_{OUT}}{R_1} = \frac{(-V_1)}{R_2} + \frac{(-V_2)}{R_4}$

$$\frac{0}{R_1} = \frac{0}{R_2} + \frac{0}{R_4} \quad \text{C.V.}$$

2. $V_1 = 0$, $V_2 = 1$ $\Rightarrow V_{OUT} = -1 \Rightarrow \frac{-1}{-R1} = \frac{0}{R_2} + \frac{(-1)}{R_4} \Rightarrow R_1 = R_4$

3. $V_1 = 1$, $V_2 = 0$ $\Rightarrow V_{OUT} = -2 \Rightarrow \frac{-2}{R1} = \frac{(-1)}{R_2} + \frac{0}{R_4} \Rightarrow R_2 \times 2 = R_1$

$$\boxed{R_1 = 2 \times R_2}$$

4. $V_1 = 1$, $V_2 = 1$ $\Rightarrow V_{OUT} = -3 \Rightarrow \frac{-3}{R_1} = \frac{-1}{R_2} + \frac{-1}{R_4}$

$$R_2 = \frac{R_1}{2}$$

$R_4 = R_1$

$$\frac{-3}{R_1} = \frac{-1}{\left(\frac{R_1}{2}\right)} + \frac{-1}{R_1}$$

$$\frac{-3}{R_1} = \frac{-2}{R_1} - \frac{1}{R_1} = \frac{-3}{R_1} \quad \text{C.V}$$

CONCLUSÃO:

$R_1 = R_4$ e $R_1 = 2 \times R_2 \Rightarrow$

$R_2 = \frac{R_1}{2}$

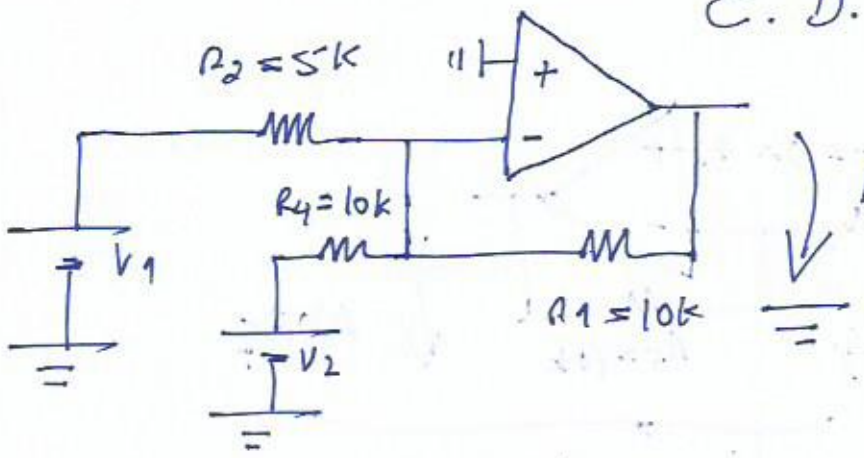
CONSIDERANDO:

$R_1 = 10k$.

$R_2 = 5k$

Circuito:

C. D. A (2 BITS)



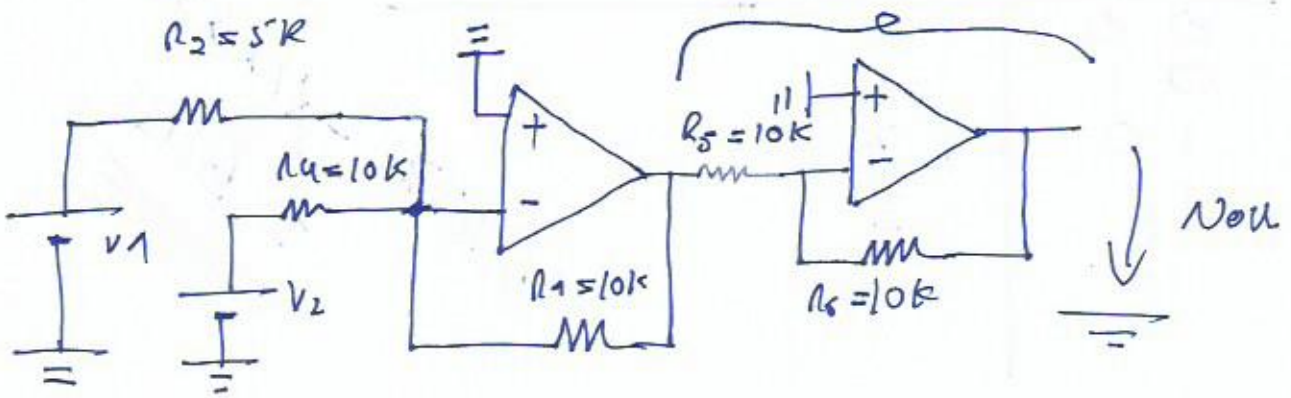
ANÁLISE CIRCUITO

$$\frac{V_1}{5K} + \frac{V_2}{10K} + \frac{V_{out}}{10K} = 0 \Rightarrow V_{out} = -\frac{10K}{5K} V_1 - \frac{10K}{10K} V_2$$

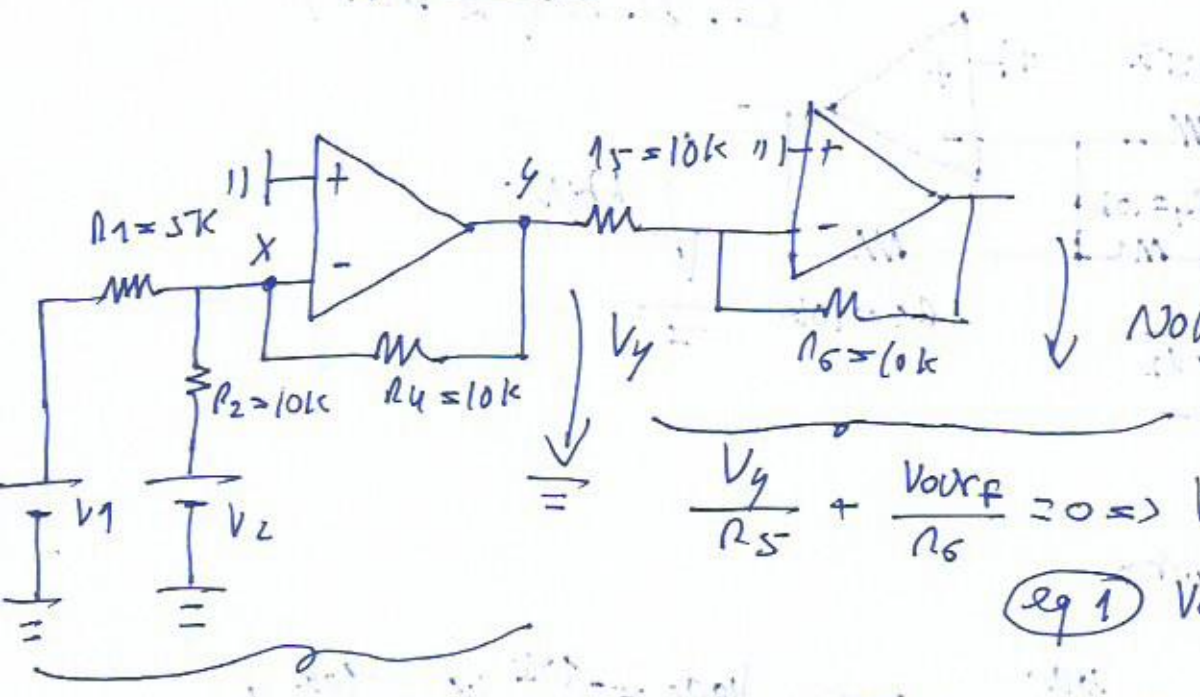
$$V_{out} = -2 \times V_1 - 1 \times V_2$$

V_1	V_2	V_{out}
0	0	0
0	1	-1
1	0	-2
1	1	-3

i) INTRODUZIR CIRCUITO INVERSOR À SAÍDA



ANÁLISE CIRCUITO



$$\frac{V_y}{R_5} + \frac{V_{out_F}}{R_6} = 0 \Rightarrow V_{out_F} = -\frac{R_6}{R_5} V_y$$

(eq 1) $V_{out_F} = -\frac{10k}{10k} V_y = -V_y$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_y}{R_4} = 0 \Rightarrow V_y = -\frac{R_4}{R_1} V_1 - \frac{R_4}{R_2} V_2$$

(eq 2) $V_y = -\frac{10k}{5k} V_1 - \frac{10k}{10k} V_2 = -V_1 \times 2 - V_2$

eq 1 → $V_{out_F} = -V_y$

eq 2 → $V_y = -2V_1 - V_2$

$\Rightarrow V_{out_F} = -(-2V_1 - V_2) = 2V_1 + V_2$

ENTRADA DIGITAL V_1 V_2	SAÍDA ANALÓGICA $V_{out_F} = 2V_1 + V_2$
0 0	0
0 1	1
1 0	2
1 1	3

DIGITAL → ANALÓGICO

CONVERSOR 2 BITS

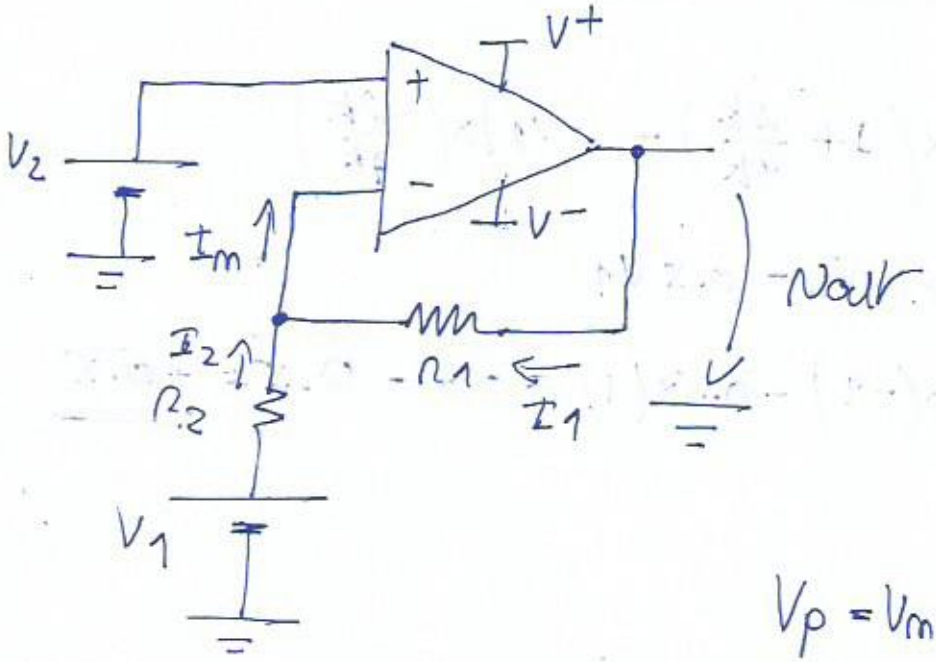
EXERCICIO 24

$V^+ = 15V$

$V^- = -15V$

$R_1 = 2k$

$R_2 = 4k$



$V_p = V_m = V_2$

$I_1 + I_2 = I_m = 0$

a) $\begin{cases} V_1 = 1V \\ V_2 = -4V \end{cases}$

$$\begin{cases} I_1 = \dfrac{V_{out} - V_m}{R_1} \\ I_2 = \dfrac{V_1 - V_m}{R_2} \end{cases}$$

$$\begin{cases} \dfrac{V_{out} - V_m}{R_1} + \dfrac{V_1 - V_m}{R_2} = 0 \\ V_m = V_2 \end{cases} \Rightarrow \frac{V_{out} - V_2}{R_1} + \frac{V_1 - V_2}{R_2} = 0$$

$$\frac{V_{out}}{R_1} = \frac{V_2}{R_1} + \frac{V_2}{R_2} - \frac{V_1}{R_2} = -\frac{V_2}{R_2} - \frac{V_1}{R_2} + \frac{V_2}{R_1}$$

$$\frac{V_{out}}{R_1} = V_2 \times \left(\frac{1}{R_2} + \frac{1}{R_1}\right) - \frac{V_1}{R_2} \Rightarrow V_{out} = V_2 \times \left(1 + \frac{R_1}{R_2}\right) - \frac{R_1}{R_2} V_1$$

$$V_{out} = V_2 \times \left(1 + \frac{R_1}{R_2}\right) + (-V_1) \times \left(\frac{R_1}{R_2}\right)$$

$R_1 = 2k$
$R_2 = 4k$

$V_1 = 1$
$V_2 = -4$

$$V_{out} = V_2 \times \left(1 + \frac{2k}{4k}\right) + (-V_1) \times \left(\frac{2k}{4k}\right)$$

$$V_{out} = 1.5 \times V_2 - 0.5\, V_1$$

$$V_{out} = 1.5 \times (-4) - 0.5 \times (1) = -6 - 0.5 = \underline{-6.5}$$

b) $V_2 = 2V$
$V_1 = 1V$

$$V_{out} = 1.5 \times V_2 - 0.5\, V_1$$
$$= 1.5 \times 2 - 0.5\,(1) = 3 - 0.5 = 2.5\,V$$

c) SIMULACA PSPICE

d) $V_2 = 3V \Rightarrow$ GANA $(V_1) \Rightarrow$ SATURACA

$$V_{out} = 1.5 \times V_2 - 0.5 \times V_1$$
$$= \left| 1.5 \times (3) - 0.5\, V_1 \right| < 15$$

$$4.5 - 0.5 \times V_1 < 15 \Rightarrow -4.5 + 0.5 V_1 > -15$$

$$4.5 - 0.5 \times V_1 > -15 \Rightarrow -4.5 + 0.5 V_1 < 15$$

$$\Rightarrow \begin{cases} 0.5 V_1 > -15 + 4.5 \\ 0.5 V_1 < 15 + 4.5 \end{cases} \Rightarrow \begin{cases} V_1 > 2 \times (-11.5) = -23\,V \\ V_1 < 2 \times 19.5 = 39\,V \end{cases}$$

$$V_1 \in \underline{[-23V, 39]\,V}$$

e)

$$V_{ou} = 1.5 \times V_2 - 0.5 \times V_1$$

$$V_1 = 2 \times \cos(2 \times \pi \times 100 \times t) = 2 \times \cos(2 \times \alpha)$$

$$\alpha = 2 \times \pi \times 50 \times t$$

$$V_2 = 2 \times (\sin(2 \times \pi \times 50 \times t))^2 = 2 \times \sin^2(\alpha)$$

Como: $\sin^2(\alpha) = 0.5 - 0.5 \times \cos(2\alpha)$

Então: $V_2 = 2 \times \sin^2(\alpha) = 2 \times (0.5 - 0.5 \times \cos(2\alpha))$

$$V_2 = 1 - \cos(2\alpha)$$

$$V_{out} = 1.5 \times (1 - \cos(2\alpha)) - 0.5 \times 2 \times \cos(2 \times \alpha)$$

$$= 1.5 - 1.5\cos(2\alpha) - \cos(2\alpha)$$

$$= 1.5 - 2.5\cos(2\alpha)$$

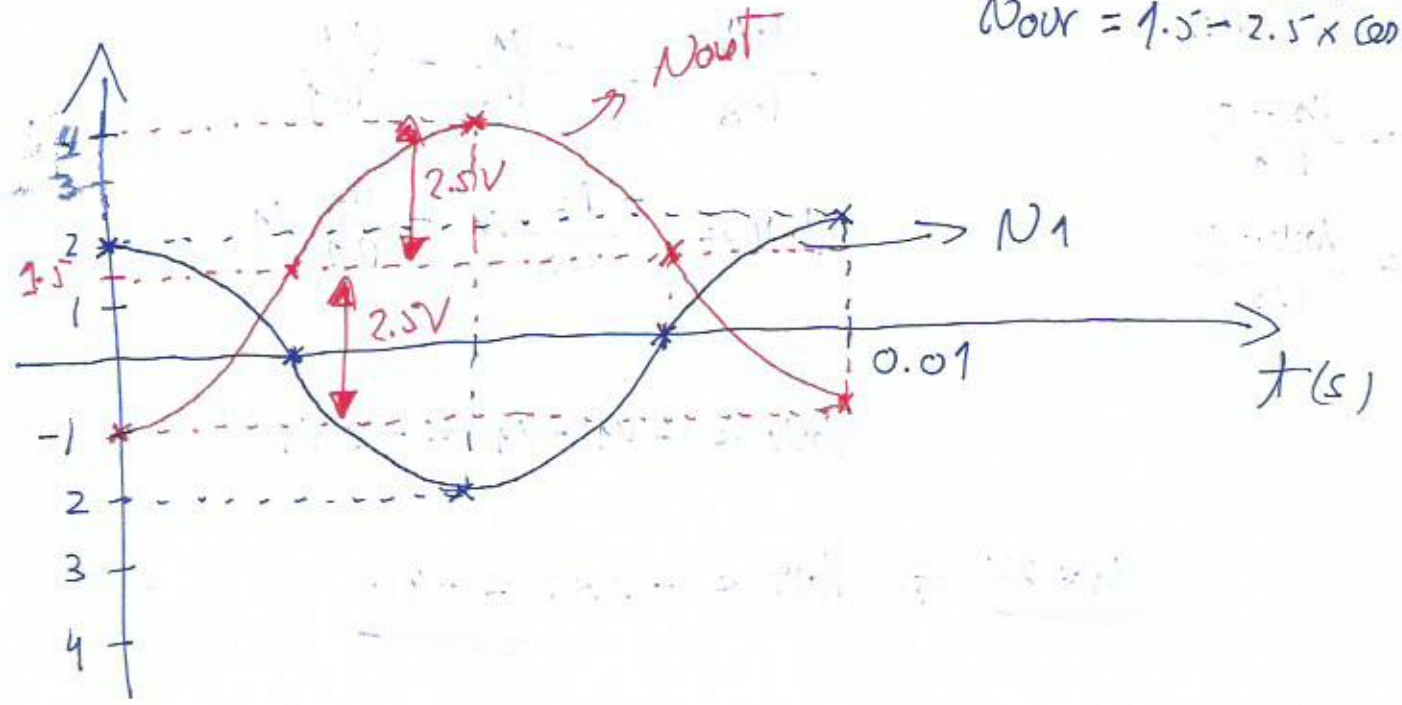
$P = 0.01s$

$f = 100$

$2\alpha = 2 \times \pi \times 100 \times t$

$V_1 = 2 \times \cos(2\alpha)$

$V_{out} = 1.5 - 2.5 \times \cos(2\alpha)$

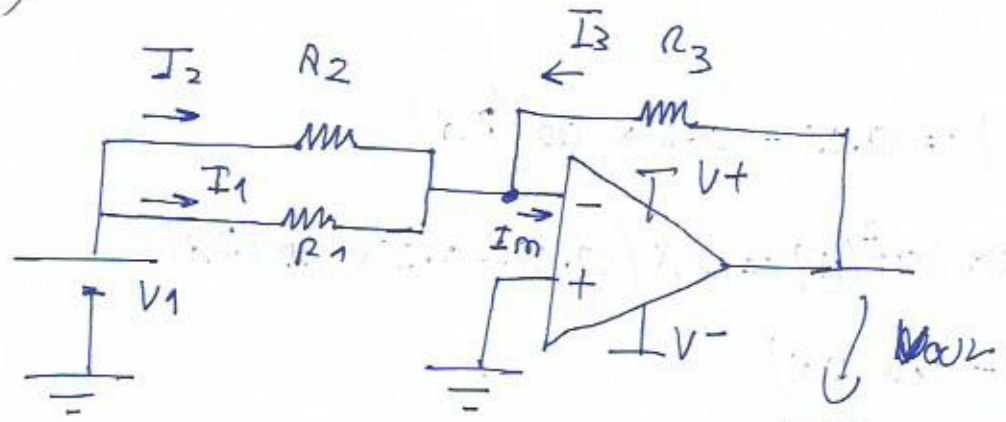


f) Simulic Python

g) Sibouly PSPICE

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25)



$R_1 = R_2 = R_3 = 1k$

$V_1 = 2V$

$V^+ = +15V$

$V^- = -15V$

a) $I_2 + I_1 + I_3 = 0 \Rightarrow \frac{V_1}{R_2} + \frac{V_1}{R_1} + \frac{V_{out}}{R_3} = 0$

$I_2 = \frac{V_1 - 0}{R_2}$

$\frac{V_{out}}{R_3} = -\frac{V_1}{R_2} - \frac{V_1}{R_1}$

$I_1 = \frac{V_1 - 0}{R_1}$

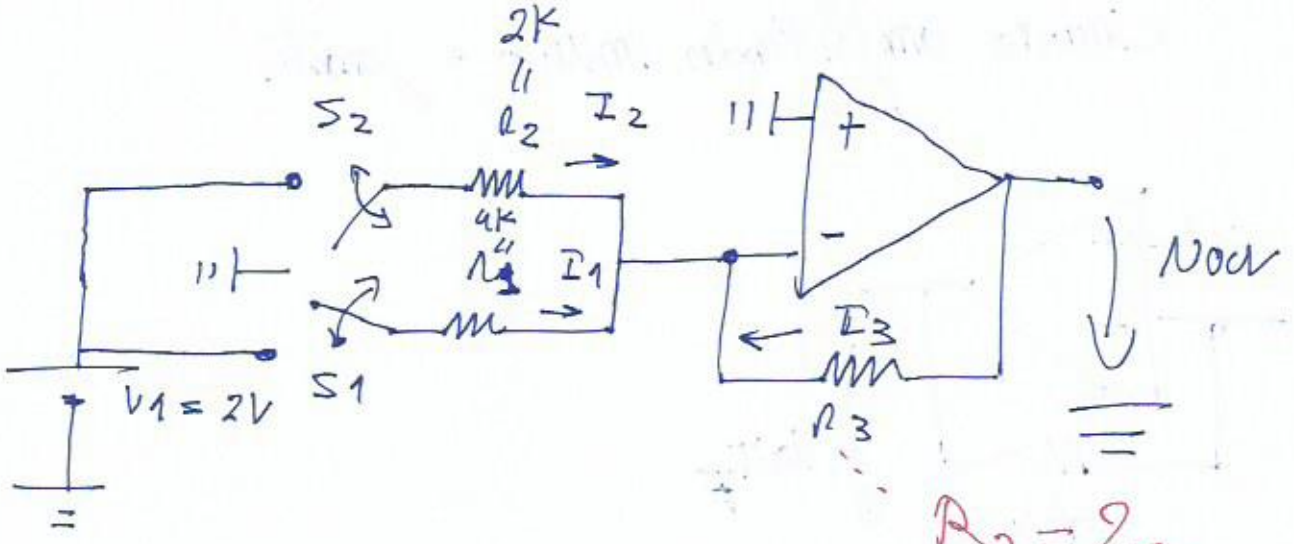
$V_{out} = -\frac{R_3}{R_2} V_1 - \frac{R_3}{R_1} V_1$

$I_3 = \frac{V_{out} - 0}{R_3}$

$V_{out} = -V_1 - V_1 = -2V_1$

$V_1 = 2V \Rightarrow V_{out} = -2 \times 2 = -4V$

25 b)



$R_3 = ?$

$$I_2 = \frac{V_1 \times S_2 - 0}{R_2} \qquad I_3 = -I_2 - I_1$$

$$I_1 = \frac{V_1 \times S_1 - 0}{R_1} \qquad \frac{V_{out}}{R_3} = -\frac{V_1 \times S_2}{R_2} - \frac{V_1 \times S_1}{R_1}$$

$$V_{out} = -\frac{R_3}{R_2} \times S_2 \times V_1 - \frac{R_3}{R_1} \times S_1 \times V_1$$

$$V_{out} = -\frac{R_3}{2k} \times S_2 \times 2 - \frac{R_3}{4k} \times S_1 \times 2$$

$$V_{out} = -\frac{R_3}{1k} \times S_2 - \frac{R_3}{2k} \times S_1$$

$R_3 = 2k$

S_2	S_1	V_{out}
0	0	0
0	1	-1
1	0	-2
1	1	-3

$V_{out} = -\frac{R_3}{1k} \times 0 - \frac{R_3}{2k} \times 0 = 0$ C.V. ←

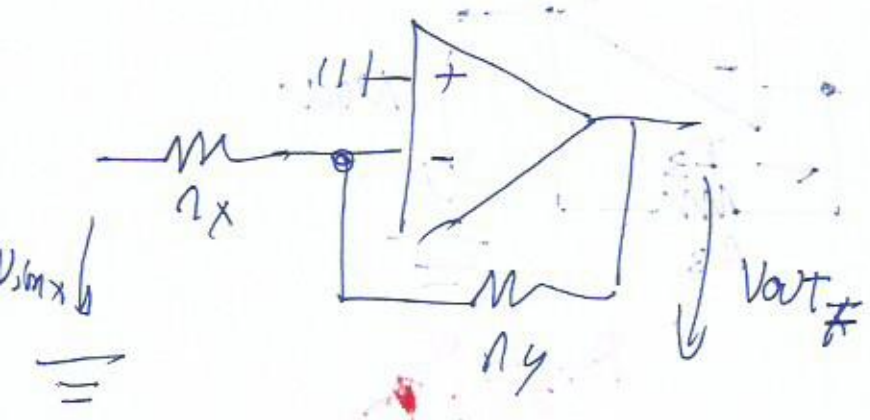
$V_{out} = -\frac{R_3}{1k} \times 0 - \frac{R_3}{2k} \times 1 = -1 \Rightarrow \boxed{R_3 = 2k}$ ←

$V_{out} = -\frac{R_3}{1k} \times 1 - \frac{R_3}{2k} \times 0 = -2 \Rightarrow \boxed{R_3 = 2k}$ ←

$V_{out} = -\frac{R_3}{1k} \times 1 - \frac{R_3}{2k} \times 1 = -3 \Rightarrow -\frac{2k}{1k} \times 1 - \frac{2k}{2k} \times 1 = -2 - 1 = -3 \Rightarrow$ C.V.

c)

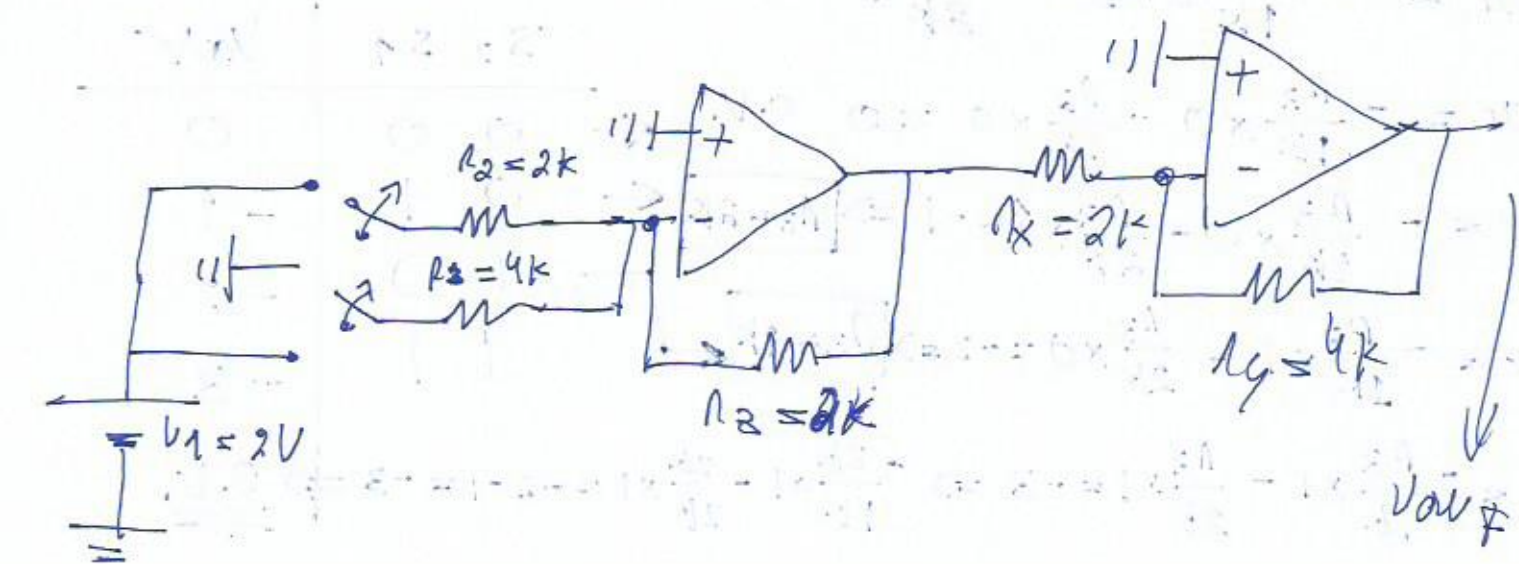
$V_{out_f} = -2 \times V_{in_x}$ => Introduzin circuito amplificador inversor + jurant.



$$\frac{V_{in_x}}{R_x} + \frac{V_{out_f}}{R_y} = 0 \Rightarrow \boxed{V_{out_f} = -\frac{R_y}{R_x} V_{in_x}}$$

$$\frac{R_y}{R_x} = 2 \Rightarrow R_y = 2 \cdot R_x \qquad R_x = 2k \qquad R_y = 4k$$

circuito final



d) A Resolução do conversor = 2V

S_2 S_1	V_{out}
0 0	0
0 1	+2
1 0	+4
1 1	+6

(2V)

e) SIMULAÇÃO PSPICE

Z6)

$R_6 = 5k \Rightarrow V_{out} = ?$

$R_1 = 5k$

$R_5 = 10k$

$R_6 = ? = 5k\,(a)$

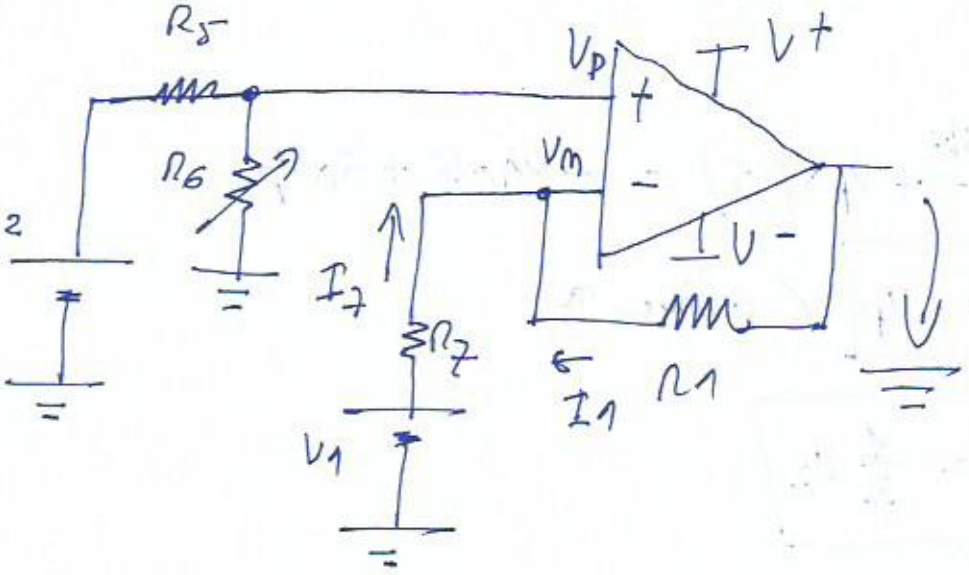
$R_7 = 1k$

$V_1 = 2$

$V_2 = 10V$

$V+ = 15V$

$V- = -15V$



$$V_p = \frac{R_6 \times V_2}{R_5 + R_6} = \frac{5k \times 10}{10k + 5k} = \frac{50k}{15k} = 3.(3)V = \frac{10}{3}$$

$$I_7 + I_1 = 0 \Rightarrow \frac{V_1 - V_p}{R_7} + \frac{V_{out} - V_p}{R_1} = 0$$

$$\frac{V_1 - 3.(3)}{1k} + \frac{V_{out} - 3.(3)}{5k} = 0 \Rightarrow \frac{V_1 - V_p}{1k} + \frac{V_{out} - V_p}{5k} = 0$$

$$\frac{V_{out}}{5k} = -\frac{V_1}{1k} + V_p \times \left(\frac{1}{1k} + \frac{1}{5k}\right) = \frac{-5k}{1k} V_1 + V_p \times \left(\frac{5k}{1k} + \frac{5k}{5k}\right)$$

$$= -5\,V_1 + V_p \times 6 = -5 \times 2 + \frac{10}{3} \times 6 =$$

b) $V_2 = 5V$

$V_{out} = ?$

$$V_p = \frac{R_6 \times V_2}{R_5 + R_6} = \frac{5k \times 5}{10k + 5k}$$

$$= \frac{25k}{15k} = 5/3$$

$$V_{out} = -5\,V_1 + 6 \times V_p$$

$$= -5 \times 2 + 6 \times 5/3 = -10 + \frac{30}{3} = -10 + 10 = \underline{0}\ V$$

c) $V_2 \Rightarrow$ GAMA SATURADA.

$$V_{out} = \frac{-V_1 \times 5k}{1k} + V_p \times (6) = -V_1 \times 5 + 6 \times V_p$$

$$\boxed{V_{out} = -5\,V_1 + 6\,V_p} \text{ (ecuação a)}$$

$$\boxed{V_p = \frac{5k \times V_2}{10k + 5k} = \frac{1}{3} V_2}$$

$$|V_{out}| = \left| -5 \times V_1 + 6 \times \frac{1}{3} V_2 \right| < 15$$

$$\left| -5 \times 2 + 2V_2 \right| < 15 \Rightarrow \begin{cases} -10 + 2V_2 < 15 \\ -10 + 2V_2 > -15 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} 2V_2 < 25 \\ 2V_2 > -5 \end{cases} \Rightarrow \begin{matrix} V_2 < \frac{25}{2} = 12.5 \\ V_2 > -\frac{5}{2} = -2.5 \end{matrix}$$

$$\boxed{V_2 \in (-2.5, 12.5)}$$

a)

$$V_{out} = -5V_1 + 6V_p$$

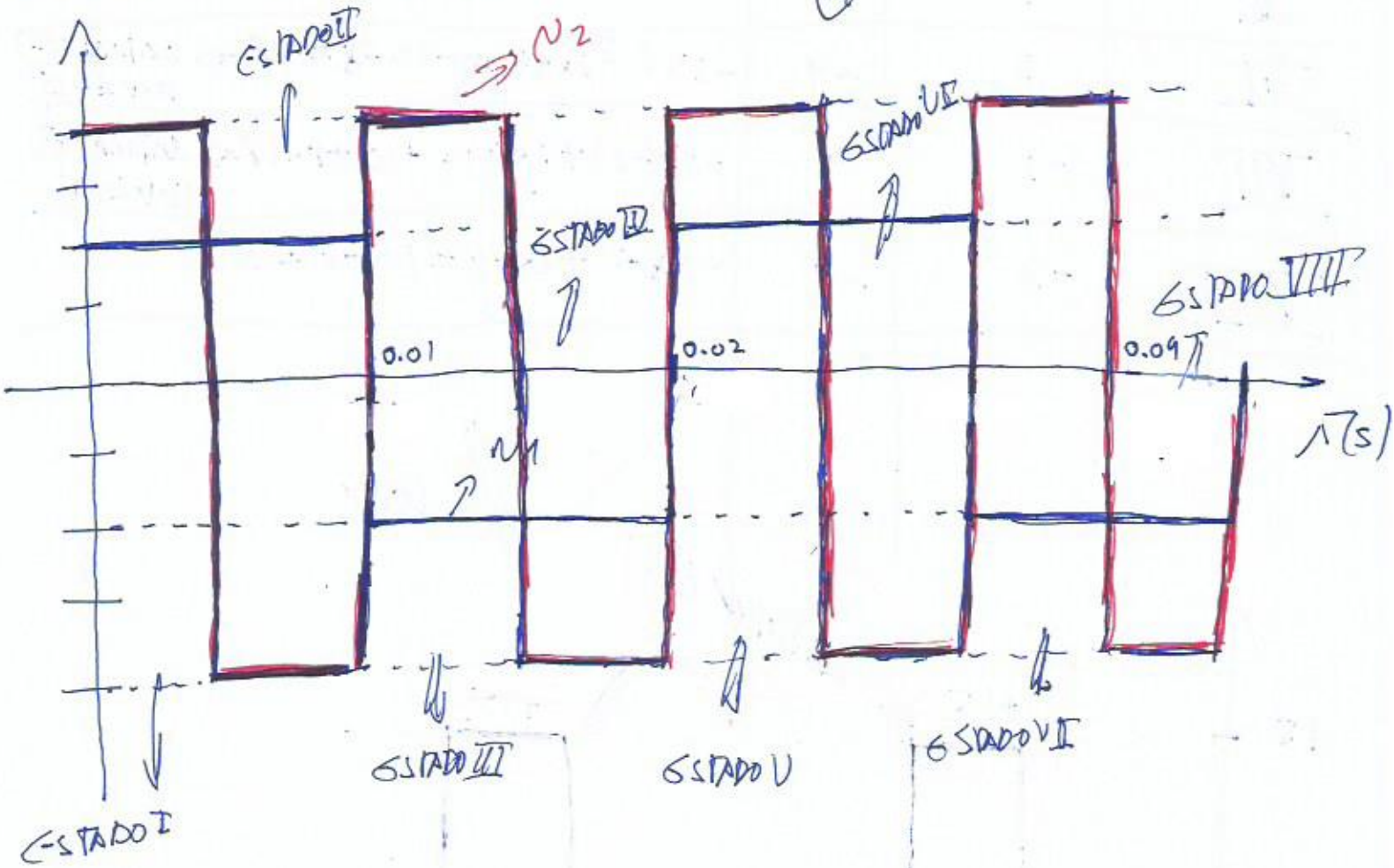
$$\doteq -5 \times V_1 + 6 \times \frac{1}{3} V_2$$

$$\boxed{V_{out} = -5 \times V_1 + 2V_2}$$

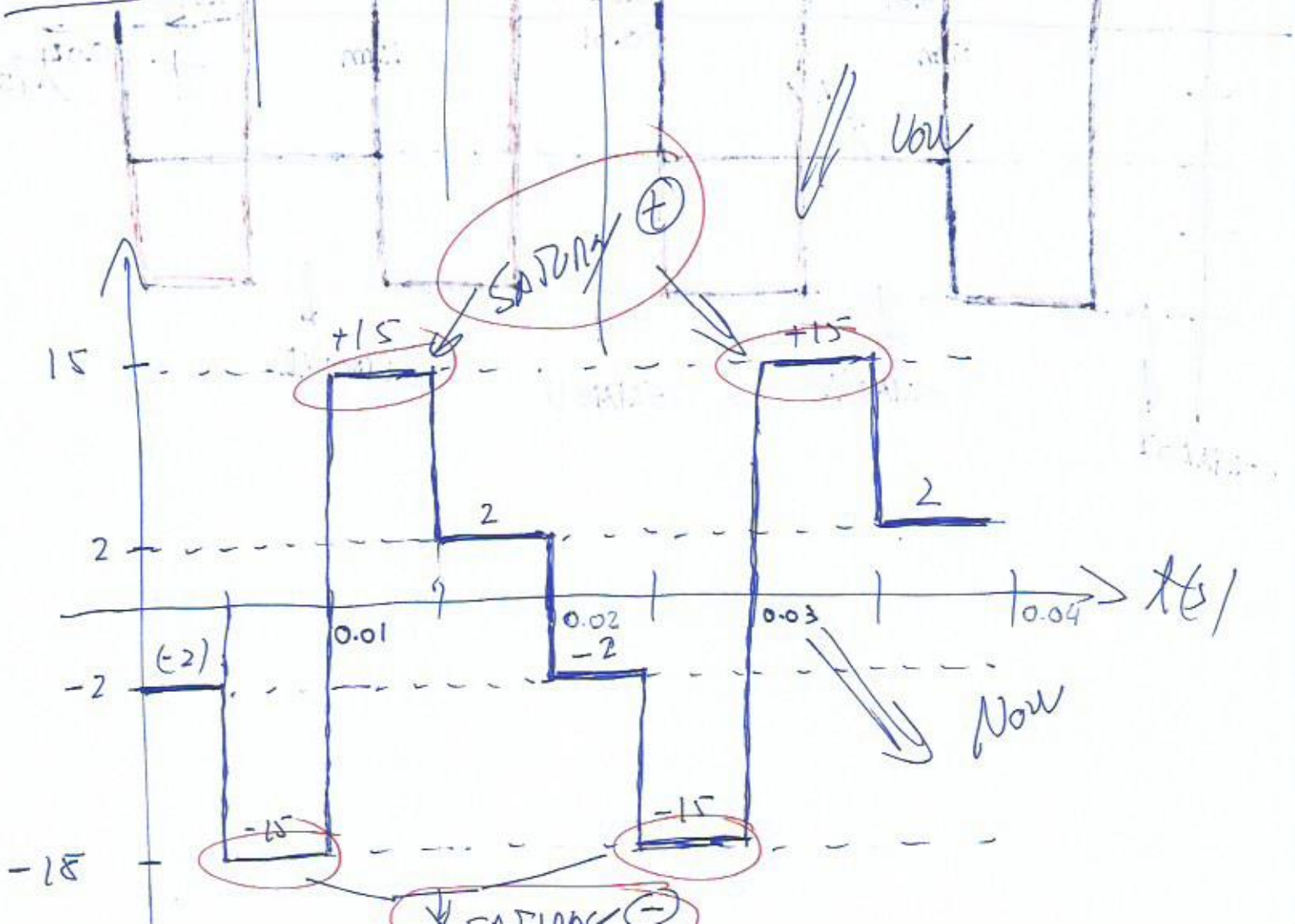
$V_1 = 2 \times \text{square}(2\pi \times 50 \times t)$

$V_2 = 4 \times \text{square}(2\pi \times 100 \times t)$

V_1 e V_2.



ESTADO	V_1	V_2	$V_{OUT} = -5V_1 + 2V_2$
I	2	4	$-5\times 2 + 2\times 4 = -10+8 = -2$
II	2	-4	$-5\times 2 + 2\times(-4) = -10-8 = -18 \Rightarrow$ SATURADO ⊖ $\Rightarrow V_{OUT} = -15$
III	-2	4	$-5\times -2 + 2\times 4 = 10+8 = 18 \Rightarrow$ SATURADO ⊕ $V_{OUT} = +15$
IV	-2	-4	$-5\times -2 + 2\times -4 = +10+(-8) = 2$
V	2	4	$-5\times 2 + 2\times 4 = -10+8 = -2$
VI	2	-4	$-5\times 2 + 2\times -4 = -10-8 = -18 \Rightarrow$ SATURADO ⊖ $V_{OUT} = -15$
VII	-2	4	$-5\times(-2) + 2\times 4 = +10+8 = 18 \Rightarrow$ SATURADO ⊕ $V_{OUT} = 15$
VIII	-2	-4	$-5\times -2 + 2\times -4 = 10-8 = 2$



9)

$$V_p = \frac{R_6 \times V_2}{R_6 + R_5} = \frac{R_6 \times 10}{R_6 + 10k}$$

$$\frac{V_1 - V_p}{R_2} + \frac{V_{OUT} - V_p}{R_1} = 0$$

$$\frac{2 - V_p}{1k} + \frac{V_{OUT} - V_p}{5'1k} = 0$$

$$\frac{V_{OUT}}{5'1k} = \frac{V_p - 2}{1k} + \frac{V_p}{5'1k} \Rightarrow V_{OUT} = \frac{5'1k}{1k}(V_p - 2) + \frac{5'1k}{5'1k} V_p$$

$$V_{OUT} = 5'1 \times (V_p - 2) + V_p = 5'1 V_p - 10 + V_p = 6'1 V_p - 10$$

$$|V_{OUT}| = \left| 6 \times \frac{R_6 \times 10}{R_6 + 10k} - 10 \right| < 15$$

$$\begin{cases} \frac{60 \times R_6}{R_6 + 10k} - 10 < 15 \\ \frac{60 \times R_6}{R_6 + 10k} - 10 > -15 \end{cases} \Rightarrow \begin{cases} \frac{60 \times R_6}{R_6 + 10k} < 25 \\ \frac{60 \times R_6}{R_6 + 10k} > -5 \end{cases}$$

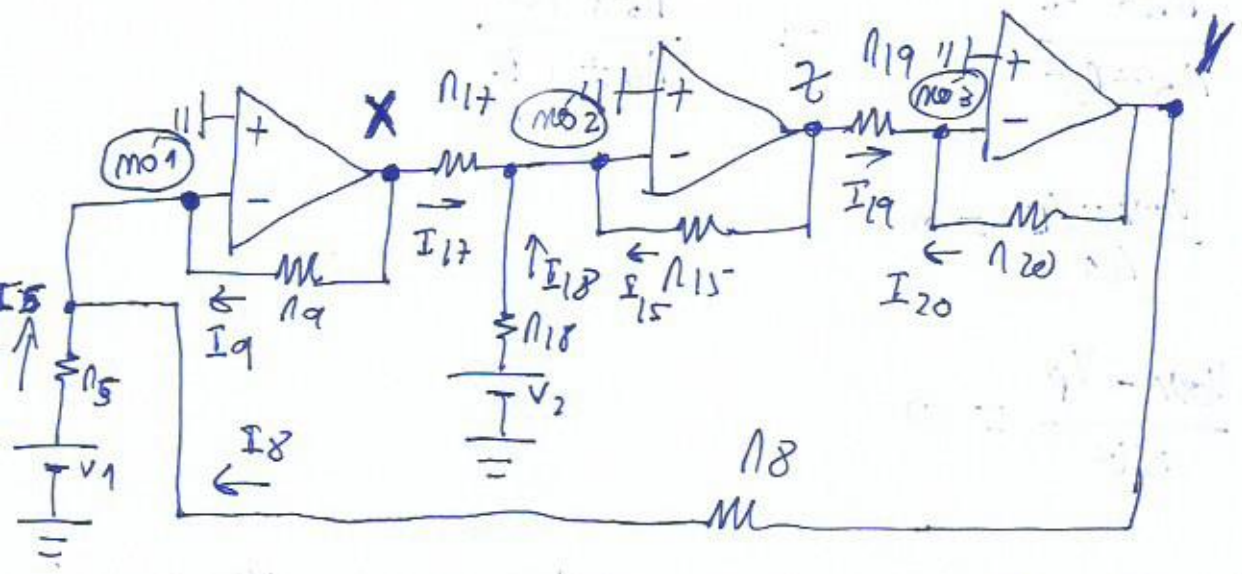
$$\begin{cases} 60 \times R_6 < (R_6 + 10k) \times 25 \\ 60 \times R_6 > -5 (R_6 + 10k) \end{cases} \Rightarrow \begin{cases} 60 R_6 - 25 R_6 < 25 \times 10k \\ 60 R_6 + 5 R_6 > -5 \times 10k \end{cases}$$

$$35 R_6 < 250k \Rightarrow R_6 < \frac{250k}{35} = 7.14k \Rightarrow \boxed{R_6 < 7k}$$

$65 R_6 > -50k \Rightarrow$ CONDICIÓ VOLTDOBLINA $R > 0$

27)

$R_5 = 2k$
$R_8 = 2k$
$R_9 = 4k$
$R_{17} = 2k$
$R_{19} = 0.5k$
$R_{15} = 2.5k$
$R_{18} = 5k$
$R_{20} = 5k$
$V_1 = -1.25$
$V_2 = 2V$



no 1: $I_5 + I_8 + I_9 = 0$

no 2: $I_{17} + I_{18} + I_{15} = 0$

no 3: $I_{19} + I_{20} = 0$

$I_5 = \frac{V_1 - 0}{R_5}$

$I_8 = \frac{V_4 - 0}{R_8}$

$I_9 = \frac{V_X - 0}{R_9}$

$I_{17} = \frac{V_X - 0}{R_{17}}$

$I_{18} = \frac{V_2 - 0}{R_{18}}$

$I_{15} = \frac{V_Z - 0}{R_{15}}$

$I_{19} = \frac{V_Z - 0}{R_{19}}$

$I_{20} = \frac{V_4 - 0}{R_{20}}$

$\frac{V_1}{R_5} + \frac{V_4}{R_8} + \frac{V_X}{R_9} = 0$

$\frac{V_X}{R_{17}} + \frac{V_2}{R_{18}} + \frac{V_Z}{R_{15}} = 0 \Rightarrow$

$\frac{V_Z - 0}{R_{19}} + \frac{V_4}{R_{20}} = 0$

$\frac{V_1}{2k} + \frac{V_4}{2k} + \frac{V_X}{4k} = 0$

$\frac{V_X}{2k} + \frac{V_2}{0.5k} + \frac{V_Z}{2.5k} = 0$

$\frac{V_Z}{5k} + \frac{V_4}{5k} = 0 \Rightarrow V_Z = -V_4$

$$\begin{cases} \frac{V_x}{4k} + \frac{V_y}{2k} = \frac{-V_1}{2k} \\ \frac{V_x}{2k} + \frac{V_z}{2.5k} = \frac{-V_2}{0.5k} \end{cases}$$

$V_z = -V_y$, $V_1 = -1.25$

$V_2 = 2V$

$$\begin{cases} \frac{V_x}{4k} + \frac{2V_y}{4k} = \frac{-2V_1}{4k} \\ \frac{5V_x}{10k} + \frac{4V_z}{10k} = \frac{-20V_2}{10k} \end{cases} \Rightarrow \begin{cases} V_x + 2V_y = -2\times(-1.25) \\ 5V_x + 4(-V_y) = -20\times(2) \end{cases}$$

$V_z = -V_y$, $V_1 = -1.25$, $V_2 = 2V$

$$\begin{cases} V_x + 2V_y = +2.5 \\ 5V_x - 4V_y = -40 \end{cases} \Rightarrow \begin{bmatrix} 1 & 2 \\ 5 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} +2.5 \\ -40 \end{bmatrix}$$

COMPUTADOR ANALÓGICO

↓

SIMULAÇÃO PSPICE

(CIRCUITO COM AMP-OPS)

COMPUTADOR DIGITAL

↓

PYTHON